NAME

mbimagecorrect − Apply brightness and contrast corrections to a single image using OpenCV algorithms.

VERSION

Version 5.0

SYNOPSIS

mbimagecorrect [**−−verbose −−help −−input=***imagefile* **−−output=***imagefile* **−−histogram−equalization −−gamma−correction=***gamma* **−−clahe−correction −−lightness−correction=***factor*]

DESCRIPTION

Mbimagecorrect applies simple brightness and contrast corrections to a single image using standard **OpenCV** algorithms. Four correction modes are available: histogram equalization, gamma correction, contrast limited adaptive histogram equalization (CLAHE), and a simple multiplicative lightness scaling. Only one correction mode is applied per run; if no correction mode is specified on the command line, CLAHE correction is used by default.

Histogram equalization converts the image from BGR to YCrCb color space, equalizes the histogram of the Y (luminance) channel only, and converts the result back to BGR. Gamma correction builds a 256 entry lookup table using the supplied gamma exponent and applies it independently to all three BGR channels of every pixel. CLAHE correction converts the image to the Lab color space, applies OpenCV's CLAHE algorithm (with a fixed clip limit of 4) to the L (lightness) channel, and converts the result back to BGR. Lightness correction converts the image to YCrCb color space and multiplies the Y channel of every pixel by a constant factor, leaving the Cr and Cb (color) channels unchanged.

The corrected image is written to the specified output file. **Mbimagecorrect** also opens two windows displaying the original and corrected images and waits for a keypress before exiting; this program is not currently useful in a non-interactive, headless batch processing context. Although the built in usage message mentions supplying an imagelist as input, the current implementation only reads a single image file named by **−−input**; it does not read or iterate over an MB-System imagelist.

Note that OpenCV enforces default maximum image dimensions when reading images (2ˆ20 pixels per side, 2ˆ30 pixels total). To work with larger images, set the environment variable **OPENCV_IO_MAX_IMAGE_PIXELS** before running the program.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−−help

This "help" flag causes the program to print out a description of its operation and then exit immediately.

−−verbose

The **−−verbose** option causes **mbimagecorrect** to print out status messages.

−−input=*imagefile*
Sets the filename of the input image to be corrected.

−−output=*imagefile*
Sets the filename of the output, corrected image.

−−histogram−equalization
Selects the histogram equalization correction mode: the Y (luminance) channel of the image (converted to YCrCb color space) is histogram equalized, and the result is converted back to BGR.

−−gamma−correction=*gamma*
Selects the gamma correction mode and sets the gamma exponent used to build the correction lookup table applied to each of the BGR channels. The default gamma value is 0.5.

−−clahe−correction
Selects the CLAHE (Contrast Limited Adaptive Histogram Equalization) correction mode: the image is converted to Lab color space, CLAHE is applied to the L (lightness) channel with a clip limit of 4, and the result is converted back to BGR. This is the default correction mode if no other correction mode option is specified.

−−lightness−correction=*factor*
Selects the simple lightness correction mode and sets the multiplicative *factor* applied to the Y (luminance) channel of the image (converted to YCrCb color space) before converting back to BGR. The default factor is 1.0.

SEE ALSO

mbsystem(1), **mbtiff2png**(1), **mbphotomosaic**(1), **mbgetphotocorrection**(1)

BUGS

Maybe. Maybe not.